Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

October 18, 2024

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Biosolids
Pace Project No.: 50384116

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on October 04, 2024. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
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CERTIFICATIONS

Project:	Biosolids
Pace Project No.:	50384116

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268
Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06
Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226
Kentucky WW Laboratory ID #: 98019
Michigan Drinking Water Laboratory #9050
Oklahoma Laboratory #: 9204
Texas Certification #: T104704355
Washington Dept of Ecology #: C1081
Wisconsin Laboratory #: 999788130
USDA Foreign Soil Permit #: 525-23-13-23119
USDA Compliance Agreement #: IN-SL-22-001

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SAMPLE SUMMARY

Project: Biosolids
Pace Project No.: 50384116

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50384116001	**Digester #3**	Solid	10/03/24 13:15	10/04/24 11:41

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SAMPLE ANALYTE COUNT

Project: Biosolids
Pace Project No.: 50384116

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50384116001	**Digester #3**	EPA 8081	KAV	21	PASI-I
		EPA 8081	CPH	8	PASI-I
		EPA 8082	BJW	8	PASI-I
		EPA 8151A	CPH	3	PASI-I
		EPA 6010	ELK	6	PASI-I
		EPA 6010	JPK	8	PASI-I
		EPA 6020	DMT	10	PASI-I
		EPA 7470	EEM	1	PASI-I
		EPA 7471	EEM	1	PASI-I
		EPA 8270	FIP	67	PASI-I
		EPA 8270	FIP	18	PASI-I
		EPA 8260	TMW	55	PASI-I
		EPA 5030/8260	SLB	13	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		SM 2540G	QAK	1	PASI-I
		EPA 9038	AEL	1	PASI-I
		EPA 9045	YAM	1	PASI-I
		EPA 351.2	MTW	1	PASI-I
		EPA 353.2	DAW	2	PASI-I
		SM 4500-NH3 G	MTW	1	PASI-I
		SM 4500-Cl-E	KCS	1	PASI-I
		SM 4500-CN-E	ATS	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8081 GCS Pesticide Solids	Analytical Method: EPA 8081 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Aldrin	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	309-00-2	
alpha-BHC	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	319-84-6	P1
beta-BHC	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	319-85-7	
delta-BHC	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	319-86-8	
gamma-BHC (Lindane)	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	58-89-9	
Chlordane (Technical)	**<12700**	ug/kg	12700	1	10/08/24 10:55	10/09/24 14:50	57-74-9	
alpha-Chlordane	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	5103-71-9	
gamma-Chlordane	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	5103-74-2	
4,4'-DDD	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	72-54-8	
4,4'-DDE	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	72-55-9	
4,4'-DDT	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	50-29-3	
Dieldrin	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	60-57-1	
Endosulfan I	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	959-98-8	
Endosulfan II	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	33213-65-9	
Endosulfan sulfate	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	1031-07-8	
Endrin	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	72-20-8	
Endrin aldehyde	**<1270**	ug/kg	1270	1	10/08/24 10:55	10/09/24 14:50	7421-93-4	
Heptachlor	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	76-44-8	
Heptachlor epoxide	**<633**	ug/kg	633	1	10/08/24 10:55	10/09/24 14:50	1024-57-3	
Toxaphene	**<12700**	ug/kg	12700	1	10/08/24 10:55	10/09/24 14:50	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	82	%.	10-139	1	10/08/24 10:55	10/09/24 14:50	2051-24-3	
8081 GCS Pest RV, TCLP	Analytical Method: EPA 8081 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24							
	Pace Analytical Services - Indianapolis							
gamma-BHC (Lindane)	**<0.00025**	mg/L	0.00025	1	10/16/24 11:45	10/16/24 17:26	58-89-9	
Chlordane (Technical)	**<0.0050**	mg/L	0.0050	1	10/16/24 11:45	10/16/24 17:26	57-74-9	
Endrin	**<0.00050**	mg/L	0.00050	1	10/16/24 11:45	10/16/24 17:26	72-20-8	
Heptachlor	**<0.00025**	mg/L	0.00025	1	10/16/24 11:45	10/16/24 17:26	76-44-8	
Heptachlor epoxide	**<0.00025**	mg/L	0.00025	1	10/16/24 11:45	10/16/24 17:26	1024-57-3	
Methoxychlor	**<0.0025**	mg/L	0.0025	1	10/16/24 11:45	10/16/24 17:26	72-43-5	
Toxaphene	**<0.0050**	mg/L	0.0050	1	10/16/24 11:45	10/16/24 17:26	8001-35-2	
Surrogates								
Decachlorobiphenyl (S)	74	%.	10-128	1	10/16/24 11:45	10/16/24 17:26	2051-24-3	
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
PCB-1016 (Aroclor 1016)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	12674-11-2	
PCB-1221 (Aroclor 1221)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	11104-28-2	
PCB-1232 (Aroclor 1232)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	11141-16-5	
PCB-1242 (Aroclor 1242)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	53469-21-9	
PCB-1248 (Aroclor 1248)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	12672-29-6	
PCB-1254 (Aroclor 1254)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	11097-69-1	
PCB-1260 (Aroclor 1260)	**<6460**	ug/kg	6460	1	10/14/24 09:56	10/15/24 15:11	11096-82-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8082 PCB Solids	Analytical Method: EPA 8082 Preparation Method: EPA 3546							
	Pace Analytical Services - Indianapolis							
Surrogates								
Tetrachloro-m-xylene (S)	81	%.	11-126	1	10/14/24 09:56	10/15/24 15:11	877-09-8	
8151A Cl Acid Herbicides TCLP	Analytical Method: EPA 8151A Preparation Method: EPA 8151							
	Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24							
	Pace Analytical Services - Indianapolis							
2,4-D	**<0.0050**	mg/L	0.0050	1	10/17/24 11:11	10/18/24 09:23	94-75-7	
2,4,5-TP (Silvex)	**<0.0050**	mg/L	0.0050	1	10/17/24 11:11	10/18/24 09:23	93-72-1	
Surrogates								
2,4-DCAA (S)	47	%.	30-170	1	10/17/24 11:11	10/18/24 09:23	19719-28-9	
6010 MET ICP	Analytical Method: EPA 6010 Preparation Method: EPA 3050							
	Pace Analytical Services - Indianapolis							
Aluminum	**6770**	mg/kg	1040	1	10/14/24 15:57	10/15/24 11:43	7429-90-5	
Calcium	**36600**	mg/kg	1040	1	10/14/24 15:57	10/16/24 14:53	7440-70-2	
Magnesium	**6680**	mg/kg	1040	1	10/14/24 15:57	10/15/24 11:43	7439-95-4	
Phosphorus	**56700**	mg/kg	2080	2	10/14/24 15:57	10/15/24 11:52	7723-14-0	N2
Potassium	**2100**	mg/kg	1040	1	10/14/24 15:57	10/15/24 11:43	7440-09-7	
Sodium	**2460**	mg/kg	1040	1	10/14/24 15:57	10/15/24 11:43	7440-23-5	
6010 MET ICP, TCLP	Analytical Method: EPA 6010 Preparation Method: EPA 3010							
	Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24							
	Pace Analytical Services - Indianapolis							
Arsenic	**<0.10**	mg/L	0.10	1	10/15/24 20:45	10/16/24 03:59	7440-38-2	
Barium	**<5.0**	mg/L	5.0	1	10/15/24 20:45	10/16/24 03:59	7440-39-3	
Cadmium	**<0.050**	mg/L	0.050	1	10/15/24 20:45	10/16/24 03:59	7440-43-9	
Chromium	**<0.10**	mg/L	0.10	1	10/15/24 20:45	10/16/24 03:59	7440-47-3	
Lead	**<0.10**	mg/L	0.10	1	10/15/24 20:45	10/16/24 03:59	7439-92-1	
Selenium	**<0.10**	mg/L	0.10	1	10/15/24 20:45	10/16/24 03:59	7782-49-2	
Silver	**<0.10**	mg/L	0.10	1	10/15/24 20:45	10/16/24 03:59	7440-22-4	
Zinc	**3.7**	mg/L	0.20	1	10/15/24 20:45	10/16/24 03:59	7440-66-6	
6020 MET ICPMS	Analytical Method: EPA 6020 Preparation Method: EPA 3050B							
	Pace Analytical Services - Indianapolis							
Arsenic	**17.6**	mg/kg	2.1	1	10/11/24 14:53	10/16/24 08:46	7440-38-2	
Cadmium	**4.1**	mg/kg	1.1	1	10/11/24 14:53	10/16/24 08:46	7440-43-9	
Chromium	**116**	mg/kg	4.2	1	10/11/24 14:53	10/16/24 08:46	7440-47-3	
Copper	**866**	mg/kg	10.5	5	10/11/24 14:53	10/17/24 03:47	7440-50-8	
Lead	**21.9**	mg/kg	10.5	5	10/11/24 14:53	10/17/24 03:47	7439-92-1	
Molybdenum	**26.7**	mg/kg	2.1	1	10/11/24 14:53	10/16/24 08:46	7439-98-7	N2
Nickel	**30.2**	mg/kg	2.1	1	10/11/24 14:53	10/16/24 08:46	7440-02-0	
Selenium	**5.7**	mg/kg	2.1	1	10/11/24 14:53	10/16/24 08:46	7782-49-2	
Silver	**3.5**	mg/kg	1.1	1	10/11/24 14:53	10/16/24 08:46	7440-22-4	
Zinc	**1630**	mg/kg	52.7	5	10/11/24 14:53	10/17/24 03:47	7440-66-6	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
7470 Mercury, TCLP	Analytical Method: EPA 7470 Preparation Method: EPA 7470 Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24 Pace Analytical Services - Indianapolis							
Mercury	**<0.0020**	mg/L	0.0020	1	10/16/24 14:49	10/17/24 11:40	7439-97-6	
7471 Mercury	Analytical Method: EPA 7471 Preparation Method: EPA 7471 Pace Analytical Services - Indianapolis							
Mercury	**<4.0**	mg/kg	4.0	1	10/14/24 13:10	10/15/24 11:26	7439-97-6	
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis							
Acenaphthene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	83-32-9	
Acenaphthylene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	208-96-8	
Anthracene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	120-12-7	
Benzo(a)anthracene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	56-55-3	
Benzo(a)pyrene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	50-32-8	
Benzo(b)fluoranthene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	205-99-2	
Benzo(g,h,i)perylene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	191-24-2	
Benzo(k)fluoranthene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	207-08-9	
Benzyl alcohol	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51	100-51-6	
4-Bromophenylphenyl ether	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	101-55-3	
Butylbenzylphthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	85-68-7	
4-Chloro-3-methylphenol	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51	59-50-7	
4-Chloroaniline	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51	106-47-8	
bis(2-Chloroethoxy)methane	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	111-91-1	
bis(2-Chloroethyl) ether	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	111-44-4	
bis(2chloro1methylethyl) ether	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	108-60-1	
2-Chloronaphthalene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	91-58-7	
2-Chlorophenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	95-57-8	
4-Chlorophenylphenyl ether	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	7005-72-3	
Chrysene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	218-01-9	
Dibenz(a,h)anthracene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	53-70-3	
Dibenzofuran	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	132-64-9	
3,3'-Dichlorobenzidine	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51	91-94-1	
2,4-Dichlorophenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	120-83-2	
Diethylphthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	84-66-2	
2,4-Dimethylphenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	105-67-9	
Dimethylphthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	131-11-3	
Di-n-butylphthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	84-74-2	
4,6-Dinitro-2-methylphenol	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51	534-52-1	
2,4-Dinitrophenol	**<34500**	ug/kg	34500	1	10/08/24 10:55	10/08/24 20:51	51-28-5	
2,4-Dinitrotoluene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	121-14-2	
2,6-Dinitrotoluene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	606-20-2	
Di-n-octylphthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	117-84-0	
bis(2-Ethylhexyl)phthalate	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	117-81-7	
Fluoranthene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	206-44-0	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 SVOC SS Soil	Analytical Method: EPA 8270 Preparation Method: EPA 3546 Pace Analytical Services - Indianapolis							
Fluorene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	86-73-7	
Hexachloro-1,3-butadiene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	87-68-3	
Hexachlorobenzene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	118-74-1	
Hexachlorocyclopentadiene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	77-47-4	
Hexachloroethane	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	67-72-1	
Indeno(1,2,3-cd)pyrene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	193-39-5	
Isophorone	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	78-59-1	
1-Methylnaphthalene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	90-12-0	
2-Methylnaphthalene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	91-57-6	
2-Methylphenol(o-Cresol)	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<14200**	ug/kg	14200	1	10/08/24 10:55	10/08/24 20:51		
Naphthalene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	91-20-3	
2-Nitroaniline	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	88-74-4	
3-Nitroaniline	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	99-09-2	
4-Nitroaniline	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	100-01-6	
Nitrobenzene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	98-95-3	
2-Nitrophenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	88-75-5	
4-Nitrophenol	**<34500**	ug/kg	34500	1	10/08/24 10:55	10/08/24 20:51	100-02-7	
N-Nitroso-di-n-propylamine	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	621-64-7	
N-Nitrosodiphenylamine	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	86-30-6	
Pentachlorophenol	**<34500**	ug/kg	34500	1	10/08/24 10:55	10/08/24 20:51	87-86-5	
Phenanthrene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	85-01-8	
Phenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	108-95-2	P1
Pyrene	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	129-00-0	
2,4,5-Trichlorophenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	95-95-4	
2,4,6-Trichlorophenol	**<7110**	ug/kg	7110	1	10/08/24 10:55	10/08/24 20:51	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	60	%.	32-98	1	10/08/24 10:55	10/08/24 20:51	4165-60-0	
Phenol-d5 (S)	73	%.	30-115	1	10/08/24 10:55	10/08/24 20:51	4165-62-2	
2-Fluorophenol (S)	62	%.	18-116	1	10/08/24 10:55	10/08/24 20:51	367-12-4	
2,4,6-Tribromophenol (S)	66	%.	15-140	1	10/08/24 10:55	10/08/24 20:51	118-79-6	
2-Fluorobiphenyl (S)	56	%.	39-97	1	10/08/24 10:55	10/08/24 20:51	321-60-8	
p-Terphenyl-d14 (S)	75	%.	49-119	1	10/08/24 10:55	10/08/24 20:51	1718-51-0	
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510 Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24 Pace Analytical Services - Indianapolis							
1,4-Dichlorobenzene	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	106-46-7	
2,4-Dinitrotoluene	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	121-14-2	
Hexachloro-1,3-butadiene	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	87-68-3	
Hexachlorobenzene	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	118-74-1	
Hexachloroethane	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	67-72-1	
2-Methylphenol(o-Cresol)	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	95-48-7	
3&4-Methylphenol(m&p Cresol)	**<0.20**	mg/L	0.20	1	10/16/24 17:01	10/17/24 12:21		
Nitrobenzene	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	98-95-3	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8270 MSSV TCLP Sep Funnel	Analytical Method: EPA 8270 Preparation Method: EPA 3510							
	Leachate Method/Date: EPA 1311; 10/14/24 18:40 Initial pH: 8.41; Final pH: 5.24							
	Pace Analytical Services - Indianapolis							
Pentachlorophenol	**<0.50**	mg/L	0.50	1	10/16/24 17:01	10/17/24 12:21	87-86-5	
Pyridine	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	110-86-1	
2,4,5-Trichlorophenol	**<0.50**	mg/L	0.50	1	10/16/24 17:01	10/17/24 12:21	95-95-4	
2,4,6-Trichlorophenol	**<0.10**	mg/L	0.10	1	10/16/24 17:01	10/17/24 12:21	88-06-2	
Surrogates								
Nitrobenzene-d5 (S)	81	%.	41-102	1	10/16/24 17:01	10/17/24 12:21	4165-60-0	
2-Fluorobiphenyl (S)	66	%.	29-84	1	10/16/24 17:01	10/17/24 12:21	321-60-8	
p-Terphenyl-d14 (S)	81	%.	54-119	1	10/16/24 17:01	10/17/24 12:21	1718-51-0	
Phenol-d5 (S)	33	%.	18-51	1	10/16/24 17:01	10/17/24 12:21	4165-62-2	
2-Fluorophenol (S)	46	%.	25-76	1	10/16/24 17:01	10/17/24 12:21	367-12-4	
2,4,6-Tribromophenol (S)	97	%.	49-147	1	10/16/24 17:01	10/17/24 12:21	118-79-6	
8260 MSV 5030 Low Level	Analytical Method: EPA 8260							
	Pace Analytical Services - Indianapolis							
Acetone	**5.0**	mg/kg	2.1	1		10/11/24 23:17	67-64-1	
Acrolein	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	107-02-8	
Acrylonitrile	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	107-13-1	
Benzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	71-43-2	
Bromodichloromethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-27-4	
Bromoform	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-25-2	
Bromomethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	74-83-9	
2-Butanone (MEK)	**1.6**	mg/kg	0.52	1		10/11/24 23:17	78-93-3	
Carbon disulfide	**<0.21**	mg/kg	0.21	1		10/11/24 23:17	75-15-0	
Carbon tetrachloride	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	56-23-5	
Chlorobenzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	108-90-7	
Chloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-00-3	
Chloroform	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	67-66-3	
Chloromethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	74-87-3	
1,2-Dibromo-3-chloropropane	**<0.21**	mg/kg	0.21	1		10/11/24 23:17	96-12-8	
Dibromochloromethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	124-48-1	
1,2-Dibromoethane (EDB)	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	106-93-4	
Dibromomethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	74-95-3	N2
1,2-Dichlorobenzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	95-50-1	
1,3-Dichlorobenzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	541-73-1	
1,4-Dichlorobenzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	106-46-7	
trans-1,4-Dichloro-2-butene	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	110-57-6	
Dichlorodifluoromethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-71-8	
1,1-Dichloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-34-3	
1,2-Dichloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	107-06-2	
1,1-Dichloroethene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-35-4	
cis-1,2-Dichloroethene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	156-59-2	
trans-1,2-Dichloroethene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	156-60-5	
1,2-Dichloropropane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	78-87-5	
cis-1,3-Dichloropropene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	10061-01-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
8260 MSV 5030 Low Level	Analytical Method: EPA 8260 Pace Analytical Services - Indianapolis							
trans-1,3-Dichloropropene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	10061-02-6	
Ethylbenzene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	100-41-4	
Ethyl methacrylate	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	97-63-2	
2-Hexanone	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	591-78-6	
Iodomethane	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	74-88-4	
Methylene Chloride	**<0.42**	mg/kg	0.42	1		10/11/24 23:17	75-09-2	
Methyl methacrylate	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	80-62-6	
4-Methyl-2-pentanone (MIBK)	**<0.52**	mg/kg	0.52	1		10/11/24 23:17	108-10-1	
Methyl-tert-butyl ether	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	1634-04-4	
Styrene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	100-42-5	
1,1,1,2-Tetrachloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	630-20-6	
1,1,2,2-Tetrachloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	79-34-5	
Tetrachloroethene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	127-18-4	
Toluene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	108-88-3	
1,1,1-Trichloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	71-55-6	
1,1,2-Trichloroethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	79-00-5	
Trichloroethene	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	79-01-6	
Trichlorofluoromethane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-69-4	
1,2,3-Trichloropropane	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	96-18-4	
Vinyl acetate	**<2.1**	mg/kg	2.1	1		10/11/24 23:17	108-05-4	
Vinyl chloride	**<0.10**	mg/kg	0.10	1		10/11/24 23:17	75-01-4	
Xylene (Total)	**<0.21**	mg/kg	0.21	1		10/11/24 23:17	1330-20-7	
Surrogates								
Dibromofluoromethane (S)	98	%.	75-135	1		10/11/24 23:17	1868-53-7	
Toluene-d8 (S)	114	%.	65-148	1		10/11/24 23:17	2037-26-5	
4-Bromofluorobenzene (S)	77	%.	63-132	1		10/11/24 23:17	460-00-4	
8260 MSV TCLP	Analytical Method: EPA 5030/8260 Leachate Method/Date: EPA 1311; 10/16/24 00:15 Pace Analytical Services - Indianapolis							
Benzene	**<0.50**	mg/L	0.50	10		10/17/24 11:07	71-43-2	
2-Butanone (MEK)	**<10.0**	mg/L	10.0	10		10/17/24 11:07	78-93-3	
Carbon tetrachloride	**<0.50**	mg/L	0.50	10		10/17/24 11:07	56-23-5	
Chlorobenzene	**<0.50**	mg/L	0.50	10		10/17/24 11:07	108-90-7	
Chloroform	**<0.50**	mg/L	0.50	10		10/17/24 11:07	67-66-3	
1,2-Dichloroethane	**<0.50**	mg/L	0.50	10		10/17/24 11:07	107-06-2	
1,1-Dichloroethene	**<0.50**	mg/L	0.50	10		10/17/24 11:07	75-35-4	
Tetrachloroethene	**<0.50**	mg/L	0.50	10		10/17/24 11:07	127-18-4	
Trichloroethene	**<0.50**	mg/L	0.50	10		10/17/24 11:07	79-01-6	
Vinyl chloride	**<0.20**	mg/L	0.20	10		10/17/24 11:07	75-01-4	
Surrogates								
4-Bromofluorobenzene (S)	96	%.	79-124	10		10/17/24 11:07	460-00-4	
Dibromofluoromethane (S)	98	%.	82-128	10		10/17/24 11:07	1868-53-7	
Toluene-d8 (S)	103	%.	73-122	10		10/17/24 11:07	2037-26-5	

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ANALYTICAL RESULTS

Project: Biosolids
Pace Project No.: 50384116

Sample: Digester #3 **Lab ID: 50384116001** Collected: 10/03/24 13:15 Received: 10/04/24 11:41 Matrix: Solid

Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.

Parameters	Results	Units	Report Limit	DF	Prepared	Analyzed	CAS No.	Qual
Percent Moisture	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Percent Moisture	**95.4**	%	0.10	1		10/10/24 09:08		N2
2540G Total Percent Solids	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Total Solids	**4.6**	%	0.10	1		10/10/24 09:08		N2
2540G Total Volatile Solids	Analytical Method: SM 2540G Pace Analytical Services - Indianapolis							
Total Volatile Solids	**50.8**	%	0.10	1		10/10/24 09:08		H1,N2
9038 Sulfate Soil	Analytical Method: EPA 9038 Preparation Method: EPA 9038 Pace Analytical Services - Indianapolis							
Sulfate	**3120**	mg/kg	2150	1	10/08/24 16:24	10/10/24 14:18	14808-79-8	N2
9045 pH Soil	Analytical Method: EPA 9045 Pace Analytical Services - Indianapolis							
pH at 25 Degrees C	**8.5**	Std. Units	0.10	1		10/15/24 14:15		H3
351.2 Total Kjeldahl Nitrogen	Analytical Method: EPA 351.2 Preparation Method: EPA 351.2 Pace Analytical Services - Indianapolis							
Nitrogen, Kjeldahl, Total	**64100**	mg/kg	5360	5	10/08/24 12:12	10/09/24 10:52	7727-37-9	N2
353.2 Nitrogen, NO2/NO3	Analytical Method: EPA 353.2 Preparation Method: EPA 353.2 Pace Analytical Services - Indianapolis							
Nitrogen, NO2 plus NO3	**<107**	mg/kg	107	1	10/11/24 23:58	10/12/24 00:55		N2
Nitrogen, Nitrate	**<107**	mg/kg	107	1	10/11/24 23:58	10/12/24 00:55	14797-55-8	N2
4500 Ammonia Soil, Distilled	Analytical Method: SM 4500-NH3 G Preparation Method: SM 4500-NH3 B Pace Analytical Services - Indianapolis							
Nitrogen, Ammonia	**29800**	mg/kg	5110	5	10/08/24 11:46	10/08/24 13:16	7664-41-7	N2
4500 Chloride in Soil	Analytical Method: SM 4500-Cl-E Preparation Method: SM 4500-Cl-E Pace Analytical Services - Indianapolis							
Chloride	**<21600**	mg/kg	21600	10	10/11/24 13:59	10/14/24 11:34	16887-00-6	D3,N2
4500CNE Cyanide, Soil	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis							
Cyanide	**<10.5**	mg/kg	10.5	1	10/15/24 10:30	10/17/24 16:22	57-12-5	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813991
QC Batch Method: EPA 7470
Analysis Method: EPA 7470
Analysis Description: 7470 Mercury TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3723476 Matrix: Water

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/L	<0.00067	0.00067	10/17/24 10:54	

LABORATORY CONTROL SAMPLE: 3723477

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	0.005	0.0049	98	80-120	

MATRIX SPIKE SAMPLE: 3723478

Parameter	Units	50384040001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.012	78	75-125	

MATRIX SPIKE SAMPLE: 3723479

Parameter	Units	50384137001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	92	75-125	

MATRIX SPIKE SAMPLE: 3723480

Parameter	Units	50384138001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	89	75-125	

MATRIX SPIKE SAMPLE: 3723481

Parameter	Units	50384139001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	91	75-125	

MATRIX SPIKE SAMPLE: 3723482

Parameter	Units	50384140001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	90	75-125	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE SAMPLE: 3723483

Parameter	Units	50384165001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Mercury	mg/L	ND	0.015	0.014	92	75-125	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3723484 3723485

Parameter	Units	50384116001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/L	<0.0020	0.015	0.015	0.014	0.014	90	90	75-125	0	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813115	Analysis Method:	EPA 7471
QC Batch Method:	EPA 7471	Analysis Description:	7471 Mercury
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3719236 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Mercury	mg/kg	<0.20	0.20	10/15/24 11:01	

LABORATORY CONTROL SAMPLE: 3719237

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Mercury	mg/kg	0.5	0.50	99	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3719238 3719239

Parameter	Units	50384365002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Mercury	mg/kg	ND	0.62	0.61	0.66	0.65	99	99	75-125	1	20	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812410
QC Batch Method: EPA 3050
Analysis Method: EPA 6010
Analysis Description: 6010 MET
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3716628 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Aluminum	mg/kg	<50.0	50.0	10/15/24 10:46	
Calcium	mg/kg	<50.0	50.0	10/15/24 10:46	
Magnesium	mg/kg	<50.0	50.0	10/15/24 10:46	
Phosphorus	mg/kg	ND	50.0	10/15/24 10:46	N2
Potassium	mg/kg	<50.0	50.0	10/15/24 10:46	
Sodium	mg/kg	<50.0	50.0	10/15/24 10:46	

LABORATORY CONTROL SAMPLE: 3716629

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Aluminum	mg/kg	500	483	97	80-120	
Calcium	mg/kg	500	496	99	80-120	
Magnesium	mg/kg	500	485	97	80-120	
Phosphorus	mg/kg	50	50.8	102	80-120	N2
Potassium	mg/kg	500	473	95	80-120	
Sodium	mg/kg	500	467	93	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3716630 3716631

Parameter	Units	50383904002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Aluminum	mg/kg	2720	511	513	5210	4700	487	386	75-125	10	20	P6
Calcium	mg/kg	84200	511	513	61400	70200	-4450	-2710	75-125	13	20	E,P6
Magnesium	mg/kg	26700	511	513	25600	26100	-200	-101	75-125	2	20	E,P6
Phosphorus	mg/kg	229	51.1	51.3	365	344	267	224	75-125	6	20	M3,N2
Potassium	mg/kg	442	511	513	1180	1310	144	169	75-125	10	20	M3
Sodium	mg/kg	491	511	513	1030	1020	106	103	75-125	1	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813834 | Analysis Method: EPA 6010
QC Batch Method: EPA 3010 | Analysis Description: 6010 MET TCLP
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3722918 Matrix: Water

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/L	<0.010	0.010	10/16/24 03:14	
Barium	mg/L	<0.50	0.50	10/16/24 03:14	
Cadmium	mg/L	<0.0050	0.0050	10/16/24 03:14	
Chromium	mg/L	<0.010	0.010	10/16/24 03:14	
Lead	mg/L	<0.010	0.010	10/16/24 03:14	
Selenium	mg/L	<0.010	0.010	10/16/24 03:14	
Silver	mg/L	<0.010	0.010	10/16/24 03:14	
Zinc	mg/L	<0.020	0.020	10/16/24 03:14	

LABORATORY CONTROL SAMPLE: 3722919

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	1	0.97	97	80-120	
Barium	mg/L	1	0.97	97	80-120	
Cadmium	mg/L	1	0.97	97	80-120	
Chromium	mg/L	1	0.97	97	80-120	
Lead	mg/L	1	0.96	96	80-120	
Selenium	mg/L	1	0.99	99	80-120	
Silver	mg/L	0.5	0.48	95	80-120	
Zinc	mg/L	1	1.0	100	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3722920 3722921

Parameter	Units	50383134001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/L	ND	10	10	10.3	10.5	102	104	50-150	2	20	
Barium	mg/L	ND	10	10	10.5	10.7	98	99	50-150	1	20	
Cadmium	mg/L	ND	10	10	10.0	10.1	100	101	50-150	1	20	
Chromium	mg/L	0.52	10	10	10.2	10.4	97	99	50-150	2	20	
Lead	mg/L	ND	10	10	9.3	9.5	93	94	50-150	1	20	
Selenium	mg/L	ND	10	10	10.3	10.4	102	104	50-150	2	20	
Silver	mg/L	ND	5	5	4.8	4.8	96	97	50-150	1	20	
Zinc	mg/L	ND	10	10	10.2	10.3	101	102	50-150	1	20	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE SAMPLE: 3722922

Parameter	Units	50383955002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.2	102	50-150	
Barium	mg/L	ND	10	11.9	100	50-150	
Cadmium	mg/L	ND	10	10.0	100	50-150	
Chromium	mg/L	ND	10	9.9	99	50-150	
Lead	mg/L	0.30	10	9.8	95	50-150	
Selenium	mg/L	ND	10	10.3	103	50-150	
Silver	mg/L	ND	5	4.9	99	50-150	
Zinc	mg/L	0.62	10	10.9	103	50-150	

MATRIX SPIKE SAMPLE: 3722923

Parameter	Units	50384040001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	11.1	111	50-150	
Barium	mg/L	ND	10	10.5	102	50-150	
Cadmium	mg/L	ND	10	10.5	105	50-150	
Chromium	mg/L	ND	10	9.9	99	50-150	
Lead	mg/L	ND	10	9.2	92	50-150	
Selenium	mg/L	ND	10	11.1	110	50-150	
Silver	mg/L	ND	5	5.4	107	50-150	
Zinc	mg/L	ND	10	10.5	104	50-150	

MATRIX SPIKE SAMPLE: 3722924

Parameter	Units	50383834001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10.1	97	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	0.52	10	10.1	96	50-150	
Lead	mg/L	ND	10	9.2	92	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	ND	10	9.9	99	50-150	

MATRIX SPIKE SAMPLE: 3722925

Parameter	Units	50384139001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10	99	50-150	
Cadmium	mg/L	ND	10	10	100	50-150	
Chromium	mg/L	ND	10	9.9	98	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	10.1	101	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE SAMPLE: 3722925

Parameter	Units	50384139001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Silver	mg/L	ND	5	4.9	98	50-150	
Zinc	mg/L	ND	10	10.3	103	50-150	

MATRIX SPIKE SAMPLE: 3722926

Parameter	Units	50384140001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	10.1	101	50-150	
Barium	mg/L	ND	10	10.1	101	50-150	
Cadmium	mg/L	ND	10	10.1	101	50-150	
Chromium	mg/L	ND	10	10	100	50-150	
Lead	mg/L	ND	10	9.6	96	50-150	
Selenium	mg/L	ND	10	10.2	102	50-150	
Silver	mg/L	ND	5	4.9	99	50-150	
Zinc	mg/L	ND	10	10.4	104	50-150	

MATRIX SPIKE SAMPLE: 3722927

Parameter	Units	50384116001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.3	103	50-150	
Barium	mg/L	<5.0	10	10.2	100	50-150	
Cadmium	mg/L	<0.050	10	10.1	101	50-150	
Chromium	mg/L	<0.10	10	9.9	99	50-150	
Lead	mg/L	<0.10	10	9.6	96	50-150	
Selenium	mg/L	<0.10	10	10.4	103	50-150	
Silver	mg/L	<0.10	5	4.9	98	50-150	
Zinc	mg/L	3.7	10	14.1	105	50-150	

MATRIX SPIKE SAMPLE: 3722928

Parameter	Units	50384651001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	<0.10	10	10.2	102	50-150	
Barium	mg/L	<5.0	10	10.5	100	50-150	
Cadmium	mg/L	<0.050	10	10.0	100	50-150	
Chromium	mg/L	<0.10	10	9.9	99	50-150	
Lead	mg/L	<0.10	10	9.6	96	50-150	
Selenium	mg/L	<0.10	10	10.3	103	50-150	
Silver	mg/L	<0.10	5	4.9	97	50-150	
Zinc	mg/L	<200 ug/L	10	10.4	103	50-150	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE SAMPLE: 3722935

Parameter	Units	50384837002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	ND	10	9.9	99	50-150	
Barium	mg/L	0.88J	10	10.5	96	50-150	
Cadmium	mg/L	ND	10	9.8	98	50-150	
Chromium	mg/L	ND	10	9.6	96	50-150	
Lead	mg/L	ND	10	9.4	94	50-150	
Selenium	mg/L	ND	10	9.9	99	50-150	
Silver	mg/L	ND	5	4.8	96	50-150	
Zinc	mg/L	ND	10	9.9	98	50-150	

MATRIX SPIKE SAMPLE: 3723233

Parameter	Units	50384942001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/L	2.6	10	12.9	103	50-150	
Barium	mg/L	7.2	10	17.3	100	50-150	
Cadmium	mg/L	ND	10	10	100	50-150	
Chromium	mg/L	ND	10	9.8	98	50-150	
Lead	mg/L	ND	10	9.5	95	50-150	
Selenium	mg/L	ND	10	10.2	102	50-150	
Silver	mg/L	ND	5	4.9	98	50-150	
Zinc	mg/L	ND	10	10.2	102	50-150	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813346
QC Batch Method: EPA 3050B
Analysis Method: EPA 6020
Analysis Description: 6020 MET
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3720530 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Arsenic	mg/kg	<0.10	0.10	10/16/24 08:29	
Cadmium	mg/kg	<0.050	0.050	10/16/24 08:29	
Chromium	mg/kg	<0.20	0.20	10/16/24 08:29	
Copper	mg/kg	<0.10	0.10	10/16/24 08:29	
Lead	mg/kg	<0.10	0.10	10/16/24 08:29	
Molybdenum	mg/kg	<0.10	0.10	10/16/24 08:29	N2
Nickel	mg/kg	<0.10	0.10	10/16/24 08:29	
Selenium	mg/kg	<0.10	0.10	10/16/24 08:29	
Silver	mg/kg	<0.050	0.050	10/16/24 08:29	
Zinc	mg/kg	<0.50	0.50	10/16/24 08:29	

LABORATORY CONTROL SAMPLE: 3720531

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Arsenic	mg/kg	4	3.9	98	80-120	
Cadmium	mg/kg	4	4.2	104	80-120	
Chromium	mg/kg	4	3.9	98	80-120	
Copper	mg/kg	4	3.9	98	80-120	
Lead	mg/kg	4	4.1	103	80-120	
Molybdenum	mg/kg	4	4.1	103	80-120	N2
Nickel	mg/kg	4	3.9	97	80-120	
Selenium	mg/kg	4	4.0	100	80-120	
Silver	mg/kg	4	4.1	103	80-120	
Zinc	mg/kg	4	3.8	94	80-120	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3720532 3720533

Parameter	Units	50384390001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Arsenic	mg/kg	1.8	3.9	4	5.7	6.2	100	109	75-125	8	20	IR
Cadmium	mg/kg	ND	3.9	4	3.5	3.6	90	90	75-125	2	20	
Chromium	mg/kg	2.9	3.9	4	5.2	6.3	61	85	75-125	18	20	M0
Copper	mg/kg	1.2	3.9	4	4.3	4.4	81	79	75-125	1	20	
Lead	mg/kg	4.2	3.9	4	11.1	9.6	178	136	75-125	14	20	M3
Molybdenum	mg/kg	ND	3.9	4	3.3	3.4	84	83	75-125	1	20	N2
Nickel	mg/kg	3.2	3.9	4	6.8	7.0	93	96	75-125	3	20	
Selenium	mg/kg	0.90	3.9	4	4.5	4.8	92	98	75-125	7	20	IR

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3720532 3720533

Parameter	Units	50384390001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Silver	mg/kg	ND	3.9	4	3.8	3.8	97	95	75-125	1	20	
Zinc	mg/kg	1.4	3.9	4	4.2	4.3	73	75	75-125	4	20	M0

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813433
QC Batch Method: EPA 8260
Analysis Method: EPA 8260
Analysis Description: 8260 MSV 5030 Low
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3721108 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1,1,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,1,1-Trichloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,1,2,2-Tetrachloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,1,2-Trichloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,1-Dichloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,1-Dichloroethene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,2,3-Trichloropropane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,2-Dibromo-3-chloropropane	mg/kg	<0.010	0.010	10/11/24 21:48	
1,2-Dibromoethane (EDB)	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,2-Dichlorobenzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,2-Dichloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,2-Dichloropropane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,3-Dichlorobenzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
1,4-Dichlorobenzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
2-Butanone (MEK)	mg/kg	<0.025	0.025	10/11/24 21:48	
2-Hexanone	mg/kg	<0.10	0.10	10/11/24 21:48	
4-Methyl-2-pentanone (MIBK)	mg/kg	<0.025	0.025	10/11/24 21:48	
Acetone	mg/kg	<0.10	0.10	10/11/24 21:48	
Acrolein	mg/kg	<0.10	0.10	10/11/24 21:48	
Acrylonitrile	mg/kg	<0.10	0.10	10/11/24 21:48	
Benzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Bromodichloromethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Bromoform	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Bromomethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Carbon disulfide	mg/kg	<0.010	0.010	10/11/24 21:48	
Carbon tetrachloride	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Chlorobenzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Chloroethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Chloroform	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Chloromethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
cis-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
cis-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Dibromochloromethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Dibromomethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	N2
Dichlorodifluoromethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Ethyl methacrylate	mg/kg	<0.10	0.10	10/11/24 21:48	
Ethylbenzene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Iodomethane	mg/kg	<0.10	0.10	10/11/24 21:48	
Methyl methacrylate	mg/kg	<0.10	0.10	10/11/24 21:48	
Methyl-tert-butyl ether	mg/kg	<0.0050	0.0050	10/11/24 21:48	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

METHOD BLANK: 3721108 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Methylene Chloride	mg/kg	<0.020	0.020	10/11/24 21:48	
Styrene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Tetrachloroethene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Toluene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
trans-1,2-Dichloroethene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
trans-1,3-Dichloropropene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
trans-1,4-Dichloro-2-butene	mg/kg	<0.10	0.10	10/11/24 21:48	
Trichloroethene	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Trichlorofluoromethane	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Vinyl acetate	mg/kg	<0.10	0.10	10/11/24 21:48	
Vinyl chloride	mg/kg	<0.0050	0.0050	10/11/24 21:48	
Xylene (Total)	mg/kg	<0.010	0.010	10/11/24 21:48	
4-Bromofluorobenzene (S)	%.	95	63-132	10/11/24 21:48	
Dibromofluoromethane (S)	%.	101	75-135	10/11/24 21:48	
Toluene-d8 (S)	%.	100	65-148	10/11/24 21:48	

LABORATORY CONTROL SAMPLE: 3721109

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1,1-Trichloroethane	mg/kg	0.05	0.046	92	67-134	
1,1,2,2-Tetrachloroethane	mg/kg	0.05	0.046	92	67-122	
1,1-Dichloroethene	mg/kg	0.05	0.046	93	57-140	
1,2-Dibromoethane (EDB)	mg/kg	0.05	0.047	94	71-126	
1,2-Dichloroethane	mg/kg	0.05	0.046	93	67-129	
1,2-Dichloropropane	mg/kg	0.05	0.044	88	71-123	
Benzene	mg/kg	0.05	0.045	90	69-125	
Chlorobenzene	mg/kg	0.05	0.045	91	68-122	
Chloroform	mg/kg	0.05	0.046	92	71-124	
cis-1,2-Dichloroethene	mg/kg	0.05	0.047	94	70-123	
Ethylbenzene	mg/kg	0.05	0.048	95	65-124	
Methyl-tert-butyl ether	mg/kg	0.05	0.046	93	69-128	
Tetrachloroethene	mg/kg	0.05	0.042	85	62-128	
Toluene	mg/kg	0.05	0.044	89	60-122	
trans-1,2-Dichloroethene	mg/kg	0.05	0.045	90	67-124	
Trichloroethene	mg/kg	0.05	0.046	92	68-128	
Vinyl chloride	mg/kg	0.05	0.051	102	52-142	
Xylene (Total)	mg/kg	0.15	0.14	91	62-122	
4-Bromofluorobenzene (S)	%.			103	63-132	
Dibromofluoromethane (S)	%.			103	75-135	
Toluene-d8 (S)	%.			100	65-148	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3721110 3721111

Parameter	Units	50384202005 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
1,1,1-Trichloroethane	mg/kg	ND	0.058	0.057	0.057	0.053	98	93	52-148	7	20	
1,1,2,2-Tetrachloroethane	mg/kg	ND	0.058	0.057	0.051	0.045	89	80	24-166	12	20	
1,1-Dichloroethene	mg/kg	ND	0.058	0.057	0.059	0.056	103	98	39-162	6	20	
1,2-Dibromoethane (EDB)	mg/kg	ND	0.058	0.057	0.057	0.052	99	91	36-141	11	20	
1,2-Dichloroethane	mg/kg	ND	0.058	0.057	0.060	0.055	103	96	48-138	9	20	
1,2-Dichloropropane	mg/kg	ND	0.058	0.057	0.055	0.050	95	89	45-140	9	20	
Benzene	mg/kg	ND	0.058	0.057	0.056	0.052	97	92	48-137	8	20	
Chlorobenzene	mg/kg	ND	0.058	0.057	0.051	0.046	88	81	28-136	10	20	
Chloroform	mg/kg	ND	0.058	0.057	0.060	0.055	103	96	54-136	9	20	
cis-1,2-Dichloroethene	mg/kg	ND	0.058	0.057	0.059	0.054	102	95	52-132	9	20	
Ethylbenzene	mg/kg	ND	0.058	0.057	0.053	0.048	93	85	24-150	10	20	
Methyl-tert-butyl ether	mg/kg	ND	0.058	0.057	0.061	0.058	105	102	57-141	5	20	
Tetrachloroethene	mg/kg	ND	0.058	0.057	0.051	0.048	89	85	26-159	7	20	
Toluene	mg/kg	24.5 ug/kg	0.058	0.057	0.085	0.077	104	92	28-150	10	20	
trans-1,2-Dichloroethene	mg/kg	ND	0.058	0.057	0.058	0.053	100	93	50-134	8	20	
Trichloroethene	mg/kg	ND	0.058	0.057	0.058	0.053	100	93	33-155	9	20	
Vinyl chloride	mg/kg	ND	0.058	0.057	0.067	0.063	116	111	37-161	7	20	
Xylene (Total)	mg/kg	ND	0.18	0.17	0.16	0.14	90	83	25-142	10	20	
4-Bromofluorobenzene (S)	%.						94	95	63-132			
Dibromofluoromethane (S)	%.						107	102	75-135			
Toluene-d8 (S)	%.						107	102	65-148			

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 814168
QC Batch Method: EPA 5030/8260
Analysis Method: EPA 5030/8260
Analysis Description: 8260 MSV TCLP
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3724261 Matrix: Water
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,1-Dichloroethene	mg/L	<0.050	0.050	10/17/24 08:48	
1,2-Dichloroethane	mg/L	<0.050	0.050	10/17/24 08:48	
2-Butanone (MEK)	mg/L	<1.0	1.0	10/17/24 08:48	
Benzene	mg/L	<0.050	0.050	10/17/24 08:48	
Carbon tetrachloride	mg/L	<0.050	0.050	10/17/24 08:48	
Chlorobenzene	mg/L	<0.050	0.050	10/17/24 08:48	
Chloroform	mg/L	<0.050	0.050	10/17/24 08:48	
Tetrachloroethene	mg/L	<0.050	0.050	10/17/24 08:48	
Trichloroethene	mg/L	<0.050	0.050	10/17/24 08:48	
Vinyl chloride	mg/L	<0.020	0.020	10/17/24 08:48	
4-Bromofluorobenzene (S)	%.	95	79-124	10/17/24 08:48	
Dibromofluoromethane (S)	%.	99	82-128	10/17/24 08:48	
Toluene-d8 (S)	%.	105	73-122	10/17/24 08:48	

LABORATORY CONTROL SAMPLE: 3724262

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	0.5	0.53	105	71-130	
1,2-Dichloroethane	mg/L	0.5	0.47	95	72-123	
2-Butanone (MEK)	mg/L	2.5	2.2	89	67-135	
Benzene	mg/L	0.5	0.52	103	76-122	
Carbon tetrachloride	mg/L	0.5	0.52	104	73-127	
Chlorobenzene	mg/L	0.5	0.55	111	76-118	
Chloroform	mg/L	0.5	0.50	100	78-121	
Tetrachloroethene	mg/L	0.5	0.60	120	71-122	
Trichloroethene	mg/L	0.5	0.53	105	74-125	
Vinyl chloride	mg/L	0.5	0.57	114	55-139	
4-Bromofluorobenzene (S)	%.			95	79-124	
Dibromofluoromethane (S)	%.			94	82-128	
Toluene-d8 (S)	%.			104	73-122	

MATRIX SPIKE SAMPLE: 3724263

Parameter	Units	50384668001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,1-Dichloroethene	mg/L	<0.025	0.5	0.58	116	53-144	
1,2-Dichloroethane	mg/L	<0.025	0.5	0.54	108	50-138	
2-Butanone (MEK)	mg/L	<0.50	2.5	2.6	104	45-138	

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE SAMPLE: 3724263

Parameter	Units	50384668001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Benzene	mg/L	<0.010	0.5	0.57	114	53-138	
Carbon tetrachloride	mg/L	<0.025	0.5	0.57	115	43-148	
Chlorobenzene	mg/L	<0.025	0.5	0.59	117	52-131	
Chloroform	mg/L	<0.025	0.5	0.56	112	54-138	
Tetrachloroethene	mg/L	<0.025	0.5	0.61	121	44-138	
Trichloroethene	mg/L	<0.025	0.5	0.57	115	49-140	
Vinyl chloride	mg/L	<0.010	0.5	0.63	126	41-147	
4-Bromofluorobenzene (S)	%.				93	79-124	
Dibromofluoromethane (S)	%.				95	82-128	
Toluene-d8 (S)	%.				103	73-122	

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812642
QC Batch Method: EPA 3546
Analysis Method: EPA 8081
Analysis Description: 8081 GCS Pesticides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3717286 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
4,4'-DDD	ug/kg	<5.0	5.0	10/09/24 15:57	
4,4'-DDE	ug/kg	<5.0	5.0	10/09/24 15:57	
4,4'-DDT	ug/kg	<5.0	5.0	10/09/24 15:57	
Aldrin	ug/kg	<2.5	2.5	10/09/24 15:57	
alpha-BHC	ug/kg	<2.5	2.5	10/09/24 15:57	
alpha-Chlordane	ug/kg	<2.5	2.5	10/09/24 15:57	
beta-BHC	ug/kg	<2.5	2.5	10/09/24 15:57	
Chlordane (Technical)	ug/kg	<50.0	50.0	10/09/24 15:57	
delta-BHC	ug/kg	<2.5	2.5	10/09/24 15:57	
Dieldrin	ug/kg	<5.0	5.0	10/09/24 15:57	
Endosulfan I	ug/kg	<2.5	2.5	10/09/24 15:57	
Endosulfan II	ug/kg	<5.0	5.0	10/09/24 15:57	
Endosulfan sulfate	ug/kg	<5.0	5.0	10/09/24 15:57	
Endrin	ug/kg	<5.0	5.0	10/09/24 15:57	
Endrin aldehyde	ug/kg	<5.0	5.0	10/09/24 15:57	
gamma-BHC (Lindane)	ug/kg	<2.5	2.5	10/09/24 15:57	
gamma-Chlordane	ug/kg	<2.5	2.5	10/09/24 15:57	
Heptachlor	ug/kg	<2.5	2.5	10/09/24 15:57	
Heptachlor epoxide	ug/kg	<2.5	2.5	10/09/24 15:57	
Toxaphene	ug/kg	<50.0	50.0	10/09/24 15:57	
Decachlorobiphenyl (S)	%.	71	10-139	10/09/24 15:57	

LABORATORY CONTROL SAMPLE: 3717287

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
4,4'-DDD	ug/kg	20	27.1	135	34-164	
4,4'-DDE	ug/kg	20	24.1	121	40-156	
4,4'-DDT	ug/kg	20	15.4	77	29-169	
Aldrin	ug/kg	10	11.9	119	41-149	
alpha-BHC	ug/kg	10	11.3	113	41-143	
alpha-Chlordane	ug/kg	10	11.2	112	39-155	
beta-BHC	ug/kg	10	10.9	109	38-157	
delta-BHC	ug/kg	10	10.3	103	26-124	
Dieldrin	ug/kg	20	25.1	125	46-155	
Endosulfan I	ug/kg	10	11.3	113	39-152	
Endosulfan II	ug/kg	20	23.0	115	40-159	
Endosulfan sulfate	ug/kg	20	20.1	101	38-150	
Endrin	ug/kg	20	23.8	119	41-162	
Endrin aldehyde	ug/kg	20	24.0	120	37-196	

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Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

LABORATORY CONTROL SAMPLE: 3717287

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
gamma-BHC (Lindane)	ug/kg	10	11.9	119	39-155	
gamma-Chlordane	ug/kg	10	11.1	111	38-162	
Heptachlor	ug/kg	10	12.0	120	45-146	
Heptachlor epoxide	ug/kg	10	11.2	112	40-158	
Decachlorobiphenyl (S)	%.			74	10-139	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717288 3717289

Parameter	Units	50383886002 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
4,4'-DDD	ug/kg	ND	20.9	21.3	18.1	22.5	87	105	28-138	21	20	R1
4,4'-DDE	ug/kg	ND	20.9	21.3	18.3	22.6	87	106	22-136	21	20	R1
4,4'-DDT	ug/kg	ND	20.9	21.3	17.9	21.8	85	102	10-164	20	20	
Aldrin	ug/kg	ND	10.5	10.7	8.8	11.3	84	106	26-137	25	20	R1
alpha-BHC	ug/kg	ND	10.5	10.7	8.9	10.2	85	96	31-139	14	20	
alpha-Chlordane	ug/kg	ND	10.5	10.7	8.7	10.7	83	100	30-139	21	20	R1
beta-BHC	ug/kg	ND	10.5	10.7	9.0	10.2	86	96	31-136	12	20	
delta-BHC	ug/kg	ND	10.5	10.7	7.8	9.8	74	92	10-149	23	20	R1
Dieldrin	ug/kg	ND	20.9	21.3	19.5	24.4	93	114	11-147	22	20	R1
Endosulfan I	ug/kg	ND	10.5	10.7	8.8	10.5	84	99	19-135	18	20	
Endosulfan II	ug/kg	ND	20.9	21.3	17.6	21.6	84	102	15-137	21	20	R1
Endosulfan sulfate	ug/kg	ND	20.9	21.3	15.3	18.6	73	87	18-128	20	20	
Endrin	ug/kg	ND	20.9	21.3	19.6	24.4	94	114	12-154	22	20	R1
Endrin aldehyde	ug/kg	ND	20.9	21.3	17.5	21.2	84	100	19-163	19	20	
gamma-BHC (Lindane)	ug/kg	ND	10.5	10.7	9.5	11.6	90	108	10-182	20	20	
gamma-Chlordane	ug/kg	ND	10.5	10.7	8.3	10.1	79	95	30-152	20	20	
Heptachlor	ug/kg	ND	10.5	10.7	10	12.4	95	116	22-158	22	20	R1
Heptachlor epoxide	ug/kg	ND	10.5	10.7	8.5	10.5	82	99	19-144	21	20	R1
Decachlorobiphenyl (S)	%.						75	62	10-139			

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813965	Analysis Method:	EPA 8081
QC Batch Method:	EPA 3510	Analysis Description:	8081 GCS TCLP Pesticides
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3723385 Matrix: Water

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chlordane (Technical)	mg/L	<0.0010	0.0010	10/16/24 16:55	
Endrin	mg/L	<0.00010	0.00010	10/16/24 16:55	
gamma-BHC (Lindane)	mg/L	<0.000050	0.000050	10/16/24 16:55	
Heptachlor	mg/L	<0.000050	0.000050	10/16/24 16:55	
Heptachlor epoxide	mg/L	<0.000050	0.000050	10/16/24 16:55	
Methoxychlor	mg/L	<0.00050	0.00050	10/16/24 16:55	
Toxaphene	mg/L	<0.0010	0.0010	10/16/24 16:55	
Decachlorobiphenyl (S)	%.	70	10-128	10/16/24 16:55	

LABORATORY CONTROL SAMPLE: 3723386

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	0.002	0.0024	121	41-162	
gamma-BHC (Lindane)	mg/L	0.001	0.0012	115	36-151	
Heptachlor	mg/L	0.001	0.0011	107	22-141	
Heptachlor epoxide	mg/L	0.001	0.0011	114	41-152	
Methoxychlor	mg/L	0.01	0.012	124	45-157	
Decachlorobiphenyl (S)	%.			70	10-128	

MATRIX SPIKE SAMPLE: 3723387

Parameter	Units	40285178001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Endrin	mg/L	<0.00050	0.01	0.0094	94	12-162	
gamma-BHC (Lindane)	mg/L	<0.00025	0.005	0.0045	91	15-156	
Heptachlor	mg/L	<0.00025	0.005	0.0043	85	11-138	
Heptachlor epoxide	mg/L	<0.00025	0.005	0.0044	87	10-164	
Methoxychlor	mg/L	<0.0025	0.05	0.049	99	10-169	
Decachlorobiphenyl (S)	%.				73	10-128	

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Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813534
QC Batch Method: EPA 3546
Analysis Method: EPA 8082
Analysis Description: 8082 PCB Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3721774 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1221 (Aroclor 1221)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1232 (Aroclor 1232)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1242 (Aroclor 1242)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1248 (Aroclor 1248)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1254 (Aroclor 1254)	ug/kg	<99.3	99.3	10/14/24 18:16	
PCB-1260 (Aroclor 1260)	ug/kg	<99.3	99.3	10/14/24 18:16	
Tetrachloro-m-xylene (S)	%.	92	11-126	10/14/24 18:16	

LABORATORY CONTROL SAMPLE: 3721775

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
PCB-1016 (Aroclor 1016)	ug/kg	331	309	93	53-125	
PCB-1260 (Aroclor 1260)	ug/kg	331	301	91	54-134	
Tetrachloro-m-xylene (S)	%.			98	11-126	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3721782 3721783

Parameter	Units	50384697001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
PCB-1016 (Aroclor 1016)	ug/kg	ND	363	370	392	386	108	104	10-170	2	20	
PCB-1260 (Aroclor 1260)	ug/kg	ND	363	370	373	412	103	111	10-156	10	20	
Tetrachloro-m-xylene (S)	%.						93	92	11-126			

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Pace Analytical Services, LLC
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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 814182
QC Batch Method: EPA 8151
Analysis Method: EPA 8151A
Analysis Description: 8151 GCS TCLP Herbicides
Laboratory: Pace Analytical Services - Indianapolis
Associated Lab Samples: 50384116001

METHOD BLANK: 3724297 Matrix: Water
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0010	0.0010	10/18/24 10:05	
2,4-D	mg/L	<0.0010	0.0010	10/18/24 10:05	
2,4-DCAA (S)	%.	94	30-170	10/18/24 10:05	

LABORATORY CONTROL SAMPLE: 3724298

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	0.005	0.0054	108	37-126	
2,4-D	mg/L	0.005	0.0049	99	28-126	
2,4-DCAA (S)	%.			93	30-170	

MATRIX SPIKE SAMPLE: 3724299

Parameter	Units	50384116001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
2,4,5-TP (Silvex)	mg/L	<0.0050	0.025	0.025	99	15-163	
2,4-D	mg/L	<0.0050	0.025	0.023	90	15-152	
2,4-DCAA (S)	%.				65	30-170	

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REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812621
QC Batch Method: EPA 3546
Analysis Method: EPA 8270
Analysis Description: 8270 Solid MSSV Microwave Short Spike
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3717220 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1-Methylnaphthalene	ug/kg	<330	330	10/08/24 15:31	
2,4,5-Trichlorophenol	ug/kg	<330	330	10/08/24 15:31	
2,4,6-Trichlorophenol	ug/kg	<330	330	10/08/24 15:31	
2,4-Dichlorophenol	ug/kg	<330	330	10/08/24 15:31	
2,4-Dimethylphenol	ug/kg	<330	330	10/08/24 15:31	
2,4-Dinitrophenol	ug/kg	<1600	1600	10/08/24 15:31	
2,4-Dinitrotoluene	ug/kg	<330	330	10/08/24 15:31	
2,6-Dinitrotoluene	ug/kg	<330	330	10/08/24 15:31	
2-Chloronaphthalene	ug/kg	<330	330	10/08/24 15:31	
2-Chlorophenol	ug/kg	<330	330	10/08/24 15:31	
2-Methylnaphthalene	ug/kg	<330	330	10/08/24 15:31	
2-Methylphenol(o-Cresol)	ug/kg	<330	330	10/08/24 15:31	
2-Nitroaniline	ug/kg	<330	330	10/08/24 15:31	
2-Nitrophenol	ug/kg	<330	330	10/08/24 15:31	
3&4-Methylphenol(m&p Cresol)	ug/kg	<660	660	10/08/24 15:31	
3,3'-Dichlorobenzidine	ug/kg	<660	660	10/08/24 15:31	
3-Nitroaniline	ug/kg	<330	330	10/08/24 15:31	
4,6-Dinitro-2-methylphenol	ug/kg	<660	660	10/08/24 15:31	
4-Bromophenylphenyl ether	ug/kg	<330	330	10/08/24 15:31	
4-Chloro-3-methylphenol	ug/kg	<660	660	10/08/24 15:31	
4-Chloroaniline	ug/kg	<660	660	10/08/24 15:31	
4-Chlorophenylphenyl ether	ug/kg	<330	330	10/08/24 15:31	
4-Nitroaniline	ug/kg	<330	330	10/08/24 15:31	
4-Nitrophenol	ug/kg	<1600	1600	10/08/24 15:31	
Acenaphthene	ug/kg	<330	330	10/08/24 15:31	
Acenaphthylene	ug/kg	<330	330	10/08/24 15:31	
Anthracene	ug/kg	<330	330	10/08/24 15:31	
Benzo(a)anthracene	ug/kg	<330	330	10/08/24 15:31	
Benzo(a)pyrene	ug/kg	<330	330	10/08/24 15:31	
Benzo(b)fluoranthene	ug/kg	<330	330	10/08/24 15:31	
Benzo(g,h,i)perylene	ug/kg	<330	330	10/08/24 15:31	
Benzo(k)fluoranthene	ug/kg	<330	330	10/08/24 15:31	
Benzyl alcohol	ug/kg	<660	660	10/08/24 15:31	
bis(2-Chloroethoxy)methane	ug/kg	<330	330	10/08/24 15:31	
bis(2-Chloroethyl) ether	ug/kg	<330	330	10/08/24 15:31	
bis(2-Ethylhexyl)phthalate	ug/kg	<330	330	10/08/24 15:31	
bis(2chloro1methylethyl) ether	ug/kg	<330	330	10/08/24 15:31	
Butylbenzylphthalate	ug/kg	<330	330	10/08/24 15:31	
Chrysene	ug/kg	<330	330	10/08/24 15:31	
Di-n-butylphthalate	ug/kg	<330	330	10/08/24 15:31	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

METHOD BLANK: 3717220 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Di-n-octylphthalate	ug/kg	<330	330	10/08/24 15:31	
Dibenz(a,h)anthracene	ug/kg	<330	330	10/08/24 15:31	
Dibenzofuran	ug/kg	<330	330	10/08/24 15:31	
Diethylphthalate	ug/kg	<330	330	10/08/24 15:31	
Dimethylphthalate	ug/kg	<330	330	10/08/24 15:31	
Fluoranthene	ug/kg	<330	330	10/08/24 15:31	
Fluorene	ug/kg	<330	330	10/08/24 15:31	
Hexachloro-1,3-butadiene	ug/kg	<330	330	10/08/24 15:31	
Hexachlorobenzene	ug/kg	<330	330	10/08/24 15:31	
Hexachlorocyclopentadiene	ug/kg	<330	330	10/08/24 15:31	
Hexachloroethane	ug/kg	<330	330	10/08/24 15:31	
Indeno(1,2,3-cd)pyrene	ug/kg	<330	330	10/08/24 15:31	
Isophorone	ug/kg	<330	330	10/08/24 15:31	
N-Nitroso-di-n-propylamine	ug/kg	<330	330	10/08/24 15:31	
N-Nitrosodiphenylamine	ug/kg	<330	330	10/08/24 15:31	
Naphthalene	ug/kg	<330	330	10/08/24 15:31	
Nitrobenzene	ug/kg	<330	330	10/08/24 15:31	
Pentachlorophenol	ug/kg	<1600	1600	10/08/24 15:31	
Phenanthrene	ug/kg	<330	330	10/08/24 15:31	
Phenol	ug/kg	<330	330	10/08/24 15:31	
Pyrene	ug/kg	<330	330	10/08/24 15:31	
2,4,6-Tribromophenol (S)	%.	81	15-140	10/08/24 15:31	
2-Fluorobiphenyl (S)	%.	78	39-97	10/08/24 15:31	
2-Fluorophenol (S)	%.	89	18-116	10/08/24 15:31	
Nitrobenzene-d5 (S)	%.	79	32-98	10/08/24 15:31	
p-Terphenyl-d14 (S)	%.	95	49-119	10/08/24 15:31	
Phenol-d5 (S)	%.	96	30-115	10/08/24 15:31	

LABORATORY CONTROL SAMPLE: 3717221

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
2,4-Dinitrotoluene	ug/kg	1670	1250	75	59-121	
2-Chlorophenol	ug/kg	1670	1140	68	55-110	
2-Methylnaphthalene	ug/kg	1670	1120	67	45-116	
4-Chloro-3-methylphenol	ug/kg	1670	1420	85	60-123	
4-Nitrophenol	ug/kg	1670	<1600	92	42-141	
Acenaphthene	ug/kg	1670	1150	69	62-105	
Acenaphthylene	ug/kg	1670	1070	64	63-107	
Anthracene	ug/kg	1670	1250	75	65-109	
Benzo(a)anthracene	ug/kg	1670	1260	76	67-114	
Benzo(a)pyrene	ug/kg	1670	1260	76	62-120	
Benzo(b)fluoranthene	ug/kg	1670	1290	77	58-125	
Benzo(g,h,i)perylene	ug/kg	1670	1290	78	57-122	
Benzo(k)fluoranthene	ug/kg	1670	1240	74	64-116	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

LABORATORY CONTROL SAMPLE: 3717221

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chrysene	ug/kg	1670	1230	74	64-116	
Dibenz(a,h)anthracene	ug/kg	1670	1230	74	58-121	
Fluoranthene	ug/kg	1670	1320	79	66-123	
Fluorene	ug/kg	1670	1170	70	66-112	
Indeno(1,2,3-cd)pyrene	ug/kg	1670	1330	80	61-121	
N-Nitroso-di-n-propylamine	ug/kg	1670	1200	72	44-106	
Naphthalene	ug/kg	1670	1080	65	56-98	
Pentachlorophenol	ug/kg	1670	<1600	56	34-128	
Phenanthrene	ug/kg	1670	1310	79	66-112	
Phenol	ug/kg	1670	1080	65	47-118	
Pyrene	ug/kg	1670	1100	66	63-116	
2,4,6-Tribromophenol (S)	%.			57	15-140	
2-Fluorobiphenyl (S)	%.			58	39-97	
2-Fluorophenol (S)	%.			62	18-116	
Nitrobenzene-d5 (S)	%.			80	32-98	
p-Terphenyl-d14 (S)	%.			77	49-119	
Phenol-d5 (S)	%.			67	30-115	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717222 3717223

Parameter	Units	50383903016 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
2,4-Dinitrotoluene	ug/kg	ND	1810	1790	1360	1110	75	62	14-127	20	20	
2-Chlorophenol	ug/kg	ND	1800	1790	1220	985	68	55	10-130	21	20	R1
2-Methylnaphthalene	ug/kg	ND	1800	1790	1310	1040	73	58	19-131	23	20	R1
4-Chloro-3-methylphenol	ug/kg	ND	1800	1790	1540	1220	85	68	18-139	23	20	R1
4-Nitrophenol	ug/kg	ND	1800	1790	1830	<1720	102	84	10-142		20	
Acenaphthene	ug/kg	ND	1800	1790	1320	1070	73	60	13-133	21	20	R1
Acenaphthylene	ug/kg	ND	1800	1790	1310	1070	73	60	17-129	20	20	
Anthracene	ug/kg	ND	1800	1790	1350	1100	75	61	18-130	21	20	R1
Benzo(a)anthracene	ug/kg	ND	1800	1790	1370	1160	76	65	19-135	16	20	
Benzo(a)pyrene	ug/kg	ND	1800	1790	1390	1150	77	64	19-133	19	20	
Benzo(b)fluoranthene	ug/kg	ND	1800	1790	1400	1190	78	66	11-142	17	20	
Benzo(g,h,i)perylene	ug/kg	ND	1800	1790	1330	1090	74	61	12-134	20	20	
Benzo(k)fluoranthene	ug/kg	ND	1800	1790	1400	1140	78	64	19-132	21	20	R1
Chrysene	ug/kg	ND	1800	1790	1420	1160	79	65	14-140	20	20	
Dibenz(a,h)anthracene	ug/kg	ND	1800	1790	1300	1030	72	57	10-146	24	20	R1
Fluoranthene	ug/kg	ND	1800	1790	1440	1160	80	65	12-149	22	20	R1
Fluorene	ug/kg	ND	1800	1790	1400	1120	78	63	17-134	22	20	R1
Indeno(1,2,3-cd)pyrene	ug/kg	ND	1800	1790	1410	1140	78	64	12-138	21	20	R1
N-Nitroso-di-n-propylamine	ug/kg	ND	1800	1790	1280	1110	71	62	17-118	14	20	
Naphthalene	ug/kg	ND	1800	1790	1220	970	67	54	17-124	22	20	R1
Pentachlorophenol	ug/kg	ND	1800	1790	<1730	<1720	59	52	10-125		20	
Phenanthrene	ug/kg	ND	1800	1790	1410	1170	78	65	14-140	18	20	
Phenol	ug/kg	ND	1800	1790	1370	1100	76	61	15-128	22	20	R1

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717222 3717223

Parameter	Units	50383903016 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Pyrene	ug/kg	ND	1800	1790	1280	1100	71	61	14-147	15	20	
2,4,6-Tribromophenol (S)	%.						57	49	15-140			
2-Fluorobiphenyl (S)	%.						62	53	39-97			
2-Fluorophenol (S)	%.						67	57	18-116			
Nitrobenzene-d5 (S)	%.						62	61	32-98			
p-Terphenyl-d14 (S)	%.						79	68	49-119			
Phenol-d5 (S)	%.						79	66	30-115			

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 814102
QC Batch Method: EPA 3510
Analysis Method: EPA 8270
Analysis Description: 8270 TCLP MSSV
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3724039 Matrix: Water

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
1,4-Dichlorobenzene	mg/L	<0.010	0.010	10/17/24 08:53	
2,4,5-Trichlorophenol	mg/L	<0.050	0.050	10/17/24 08:53	
2,4,6-Trichlorophenol	mg/L	<0.010	0.010	10/17/24 08:53	
2,4-Dinitrotoluene	mg/L	<0.010	0.010	10/17/24 08:53	
2-Methylphenol(o-Cresol)	mg/L	<0.010	0.010	10/17/24 08:53	
3&4-Methylphenol(m&p Cresol)	mg/L	<0.020	0.020	10/17/24 08:53	
Hexachloro-1,3-butadiene	mg/L	<0.010	0.010	10/17/24 08:53	
Hexachlorobenzene	mg/L	<0.010	0.010	10/17/24 08:53	
Hexachloroethane	mg/L	<0.010	0.010	10/17/24 08:53	
Nitrobenzene	mg/L	<0.010	0.010	10/17/24 08:53	
Pentachlorophenol	mg/L	<0.050	0.050	10/17/24 08:53	
Pyridine	mg/L	<0.010	0.010	10/17/24 08:53	
2,4,6-Tribromophenol (S)	%.	73	49-147	10/17/24 08:53	
2-Fluorobiphenyl (S)	%.	53	29-84	10/17/24 08:53	
2-Fluorophenol (S)	%.	36	25-76	10/17/24 08:53	
Nitrobenzene-d5 (S)	%.	67	41-102	10/17/24 08:53	
p-Terphenyl-d14 (S)	%.	81	54-119	10/17/24 08:53	
Phenol-d5 (S)	%.	26	18-51	10/17/24 08:53	

LABORATORY CONTROL SAMPLE: 3724040

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	0.05	0.032	64	15-85	
2,4,5-Trichlorophenol	mg/L	0.05	<0.050	80	47-116	
2,4,6-Trichlorophenol	mg/L	0.05	0.039	78	48-115	
2,4-Dinitrotoluene	mg/L	0.05	0.042	83	44-112	
2-Methylphenol(o-Cresol)	mg/L	0.05	0.035	69	39-98	
3&4-Methylphenol(m&p Cresol)	mg/L	0.1	0.062	62	35-92	
Hexachloro-1,3-butadiene	mg/L	0.05	0.031	61	10-81	
Hexachlorobenzene	mg/L	0.05	0.032	64	35-98	
Hexachloroethane	mg/L	0.05	0.033	66	10-83	
Nitrobenzene	mg/L	0.05	0.042	84	45-110	
Pentachlorophenol	mg/L	0.05	<0.050	59	24-127	
Pyridine	mg/L	0.05	0.026	53	10-83	
2,4,6-Tribromophenol (S)	%.			89	49-147	
2-Fluorobiphenyl (S)	%.			67	29-84	
2-Fluorophenol (S)	%.			49	25-76	
Nitrobenzene-d5 (S)	%.			82	41-102	
p-Terphenyl-d14 (S)	%.			93	54-119	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

LABORATORY CONTROL SAMPLE: 3724040

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Phenol-d5 (S)	%.			33	18-51	

MATRIX SPIKE SAMPLE: 3724041

Parameter	Units	50384136001 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
1,4-Dichlorobenzene	mg/L	ND	0.5	0.29	59	17-87	
2,4,5-Trichlorophenol	mg/L	ND	0.5	<0.50	65	43-126	
2,4,6-Trichlorophenol	mg/L	ND	0.5	0.32	63	43-123	
2,4-Dinitrotoluene	mg/L	ND	0.5	0.12	23	48-106	M1
2-Methylphenol(o-Cresol)	mg/L	ND	0.5	0.18	35	38-100	M1
3&4-Methylphenol(m&p Cresol)	mg/L	ND	1	0.39	39	31-98	
Hexachloro-1,3-butadiene	mg/L	ND	0.5	0.34	67	10-90	
Hexachlorobenzene	mg/L	ND	0.5	0.24	48	32-94	
Hexachloroethane	mg/L	ND	0.5	0.33	67	10-88	
Nitrobenzene	mg/L	ND	0.5	0.39	77	46-105	
Pentachlorophenol	mg/L	ND	0.5	<0.50	55	20-141	
Pyridine	mg/L	ND	0.5	<0.10	19	10-85	
2,4,6-Tribromophenol (S)	%.				62	49-147	
2-Fluorobiphenyl (S)	%.				67	29-84	
2-Fluorophenol (S)	%.				37	25-76	
Nitrobenzene-d5 (S)	%.				78	41-102	
p-Terphenyl-d14 (S)	%.				66	54-119	
Phenol-d5 (S)	%.				30	18-51	

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813084	Analysis Method:	SM 2540G
QC Batch Method:	SM 2540G	Analysis Description:	Dry Weight/Percent Moisture
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

SAMPLE DUPLICATE: 3719142

Parameter	Units	50384116001 Result	Dup Result	RPD	Max RPD	Qualifiers
Percent Moisture	%	95.4	95.4	0	10	N2

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813085
QC Batch Method: SM 2540G
Analysis Method: SM 2540G
Analysis Description: 2540G Total Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

SAMPLE DUPLICATE: 3719144

Parameter	Units	50384116001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Solids	%	4.6	4.6	1	10	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813073 | Analysis Method: SM 2540G
QC Batch Method: SM 2540G | Analysis Description: 2540G Total Volatile Solids
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

SAMPLE DUPLICATE: 3719114

Parameter	Units	50384116001 Result	Dup Result	RPD	Max RPD	Qualifiers
Total Volatile Solids	%	50.8	51.9	2	10	H1,N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812749
QC Batch Method: EPA 9038
Analysis Method: EPA 9038
Analysis Description: 9038 S2ulfate Soil
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3717715 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Sulfate	mg/kg	<100	100	10/10/24 14:15	N2

LABORATORY CONTROL SAMPLE: 3717716

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Sulfate	mg/kg	200	188	94	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717717 3717718

Parameter	Units	50384116001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Sulfate	mg/kg	3120	4310	4310	8540	8400	126	123	90-110	2	20	M3,N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813704	Analysis Method:	EPA 9045
QC Batch Method:	EPA 9045	Analysis Description:	9045 pH
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

SAMPLE DUPLICATE: 3722328

Parameter	Units	50384298001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	8.4	8.4	0	2	H3

SAMPLE DUPLICATE: 3722329

Parameter	Units	50384410001 Result	Dup Result	RPD	Max RPD	Qualifiers
pH at 25 Degrees C	Std. Units	7.8	7.8	1	2	H3

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812660
QC Batch Method: EPA 351.2
Analysis Method: EPA 351.2
Analysis Description: 351.2 TKN
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3717320 Matrix: Solid

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	<49.8	49.8	10/09/24 09:07	N2

LABORATORY CONTROL SAMPLE: 3717321

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Kjeldahl, Total	mg/kg	493	467	95	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717322 3717323

Parameter	Units	50383920001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Kjeldahl, Total	mg/kg	39700	2840	2930	42900	42000	110	78	90-110	2	20	M0,N2

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813449	Analysis Method:	EPA 353.2
QC Batch Method:	EPA 353.2	Analysis Description:	353.2 Nitrate + Nitrite
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3721322 Matrix: Solid

Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Nitrate	mg/kg	<5.0	5.0	10/12/24 00:49	N2
Nitrogen, NO2 plus NO3	mg/kg	<5.0	5.0	10/12/24 00:49	N2

LABORATORY CONTROL SAMPLE: 3721323

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Nitrate	mg/kg	9.9	10.1	102	90-110	N2
Nitrogen, NO2 plus NO3	mg/kg	19.8	20.3	102	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3721324 3721325

Parameter	Units	50384631001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Nitrate	mg/kg	ND	49	49	45.4	45.2	92	92	90-110	0	20	N2
Nitrogen, NO2 plus NO3	mg/kg	ND	97.5	97.5	95.1	95.2	97	97	90-110	0	20	N2

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QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 812624
QC Batch Method: SM 4500-NH3 B
Analysis Method: SM 4500-NH3 G
Analysis Description: 4500 Ammonia, Distilled
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3717232   Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Nitrogen, Ammonia	mg/kg	<5.0	5.0	10/08/24 12:48	N2

LABORATORY CONTROL SAMPLE: 3717233

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Nitrogen, Ammonia	mg/kg	19.8	20.3	103	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3717234   3717235

Parameter	Units	50384189001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Nitrogen, Ammonia	mg/kg	9990	1020	1030	10800	11100	80	111	90-110	3	20	M3,N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch: 813415
QC Batch Method: SM 4500-Cl-E
Analysis Method: SM 4500-Cl-E
Analysis Description: 4500 Chloride
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3720997 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Chloride	mg/kg	<99.7	99.7	10/14/24 11:32	N2

LABORATORY CONTROL SAMPLE: 3720998

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Chloride	mg/kg	199	210	105	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3720999 3721000

Parameter	Units	50384448001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Chloride	mg/kg	<1010	2010	2010	2130	2170	103	105	90-110	2	20	N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

QUALITY CONTROL DATA

Project: Biosolids
Pace Project No.: 50384116

QC Batch:	813720	Analysis Method:	SM 4500-CN-E
QC Batch Method:	SM 4500-CN-C	Analysis Description:	4500CNE Cyanide
		Laboratory:	Pace Analytical Services - Indianapolis

Associated Lab Samples: 50384116001

METHOD BLANK: 3722348 Matrix: Solid
Associated Lab Samples: 50384116001

Parameter	Units	Blank Result	Reporting Limit	Analyzed	Qualifiers
Cyanide	mg/kg	<0.48	0.48	10/17/24 16:20	N2

LABORATORY CONTROL SAMPLE: 3722349

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/kg	5	4.9	98	90-110	N2

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 3722350 3722351

Parameter	Units	50384116001 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qual
Cyanide	mg/kg	<10.5	108	108	98.9	91.3	90	83	90-110	8	20	M0,N2

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

QUALIFIERS

Project: Biosolids
Pace Project No.: 50384116

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

ANALYTE QUALIFIERS

D3	Sample was diluted due to the presence of high levels of non-target analytes or other matrix interference.
E	Analyte concentration exceeded the calibration range. The reported result is estimated.
H1	Analysis conducted outside the recognized method holding time.
H3	Sample was received or analysis requested beyond the recognized method holding time.
IR	The internal standard recovery associated with this result exceeds the upper control limit. The reported result should be considered an estimated value.
M0	Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.
M1	Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery.
M3	Matrix spike recovery was outside laboratory control limits due to matrix interferences.
N2	The lab does not hold NELAC/TNI accreditation for this parameter but other accreditations/certifications may apply. A complete list of accreditations/certifications is available upon request.
P1	Routine initial sample volume or weight was not used for extraction, resulting in elevated reporting limits.
P6	Matrix spike recovery was outside laboratory control limits due to a parent sample concentration notably higher than the spike level.
R1	RPD value was outside control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Biosolids
Pace Project No.: 50384116

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50384116001	**Digester #3**	EPA 3546	812642	EPA 8081	812762
50384116001	**Digester #3**	EPA 3510	813965	EPA 8081	814100
50384116001	**Digester #3**	EPA 3546	813534	EPA 8082	813605
50384116001	**Digester #3**	EPA 8151	814182	EPA 8151A	814401
50384116001	**Digester #3**	EPA 3050	812410	EPA 6010	813784
50384116001	**Digester #3**	EPA 3010	813834	EPA 6010	813920
50384116001	**Digester #3**	EPA 3050B	813346	EPA 6020	813522
50384116001	**Digester #3**	EPA 7470	813991	EPA 7470	814207
50384116001	**Digester #3**	EPA 7471	813115	EPA 7471	813716
50384116001	**Digester #3**	EPA 3546	812621	EPA 8270	812716
50384116001	**Digester #3**	EPA 3510	814102	EPA 8270	814169
50384116001	**Digester #3**	EPA 8260	813433		
50384116001	**Digester #3**	EPA 5030/8260	814168		
50384116001	**Digester #3**	SM 2540G	813084		
50384116001	**Digester #3**	SM 2540G	813085		
50384116001	**Digester #3**	SM 2540G	813073		
50384116001	**Digester #3**	EPA 9038	812749	EPA 9038	813166
50384116001	**Digester #3**	EPA 9045	813704		
50384116001	**Digester #3**	EPA 351.2	812660	EPA 351.2	812826
50384116001	**Digester #3**	EPA 353.2	813449	EPA 353.2	813450
50384116001	**Digester #3**	SM 4500-NH3 B	812624	SM 4500-NH3 G	812685
50384116001	**Digester #3**	SM 4500-Cl-E	813415	SM 4500-Cl-E	813550
50384116001	**Digester #3**	SM 4500-CN-C	813720	SM 4500-CN-E	814345

Pace®

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY- Affix Workorder/Login Label Here

WO#: 50384116

50384116

Company Name:	City of Cadillac	Contact/Report To:	Cindy Tomaszewski
Street Address:	1121 Plett Rd, Cadillac, MI 49601	Phone #:	231-775-2368
		E-Mail:	ctomaszewski@cadillac-mi.net
		Cc E-Mail:	
Customer Project #:			
Project Name:	Biosolids - FALL	Invoice To:	Cindy Tomaszewski
		Invoice E-Mail:	ctomaszewski@cadillac-mi.net
Site Collection Info/Facility ID (as applicable):		Purchase Order # (if applicable):	
		Quote #:	
Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [X] ET		County / State origin of sample(s):	Michigan

Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: NPDES — Reportable [ ] Yes [ ] No

Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ______

DW PWSID # or WW Permit # as applicable:

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [X] No — Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW),Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **: 10 | 10 | 10 | 10

Identify Container Preservative Type***: 1 | 1 | 1 | 1

Analysis Requested

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	8260 MSV 5030 VOC	8270 SVOC; 8081 Pest; 8082 PCB	Metals; Wet Chem	TCLP - VOC/SV/Met/Pest/Herb	Sample Comment
DIGESTER #3	BS	G	—	—	10-3-24	1315	4	—	-	X	X	X	X	

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only:
Proj. Mgr: **Brian Hall**
AcctNum / Client ID:
Table #:
Profile / Template: **8311**
Prelog / Bottle Ord. ID: **EZ 3084196**

Preservation non-conformance identified for sample.

Additional Instructions from Pace®:
Wet Chem: TKN,NH3,CN,Cl,NO2/NO3,SO4,pH,%Solid,TVS
17 Metals: 6010/6020/7471

Collected By: (Printed Name) TAYSHA LACZKO
Signature: T. Laczko

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
T. Laczko / City of Cadillac	10-3-24 / 1327	Ship VIA UPS Ground	
		D. [illegible] / Pace	10/4 1141

Tracking Number:

Delivered by: [ ] In-Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Pace

Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 10/4 1250	**Evaluated By:** DAS	**WO# : 50384116** PM: BJH Due Date: 10/18/24 CLIENT: GR-Cadillac	
Client: City of Cadillac	**PM:** BJH		
Lab Notified of Rush or Short Holds: YES NO NA			

Project Received Via: FedEx UPS Client Pace Courier Other: ____________				**Comments:**
Custody Seal Present and Intact:	YES	NO	NA	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA	
Short Hold Present (≤ 48 Hours):	YES	NO		
Sample Received in Hold:	YES	NO		
Custody Signature Present:	YES	NO		
Collector Signature Present:	YES	NO		
Sample Collected Today and On Ice:	YES	NO	N/A	
IR Gun #: 350 351 **Therm #:** 282 283	**Temp. should be 0°C - 6°C (Initial/Corrected)**			
Ice Type: WET Bagged / WET Loose BLUE NONE	**1.** Cooler Temp. Upon Receipt: 1.7/1.5 °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED	**2.** Cooler Temp. Upon Receipt: ____________ °C			
Temp Blank Received:	YES	NO		
Sample Label Matches COC (ID/Date/Time):	YES	NO		
Container Intact:	YES	NO		
Correct Container:	YES	NO		
Sufficient Volume:	YES	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. **pH Strip Lot #:** ____________ *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	YES	NO	N/A	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** ____________ *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	YES	NO	N/A	
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A	
Trip Blank Received: HCl MeOH Other: ____________	YES	NO	ON HOLD	

Comments:	**3.** Cooler Temp. Upon Receipt: ____________ °C
	4. Cooler Temp. Upon Receipt: ____________ °C
	Non-Conformance Form Required: YES NO